Name: ____________________

New York State Testing Program

Mathematics Test Session 1

Grade 6

Spring 2026

RELEASED QUESTIONS

Grade 6 Mathematics Reference Sheet

CONVERSIONS

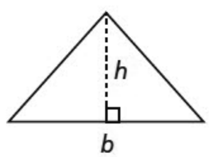
1 yard = 3 feet
1 mile = 5,280 feet

1 cup = 8 fluid ounces
1 pint = 2 cups
1 quart = 2 pints
1 gallon = 4 quarts
1 liter = 1,000 milliliters

1 pound = 16 ounces
1 ton = 2,000 pounds
1 kilogram = 1,000 grams

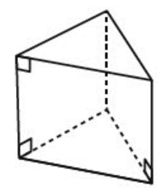
FORMULAS AND FIGURES

Triangle

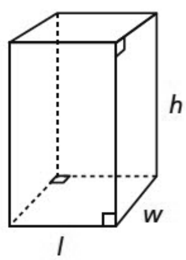


$A = \frac{1}{2}bh$

Right Triangular Prism

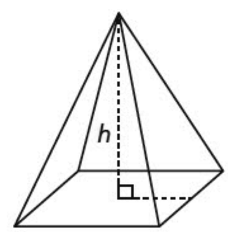


Right Rectangular Prism



$V = lwh$
$V = Bh$

Right Rectangular Pyramid



Session 1

TIPS FOR TAKING THE TEST

Here are some ideas to help you do your best:

- Read each question carefully. Take your time.
- You have a ruler, a protractor, and a reference sheet that you can use on the test if they help you answer the question.

1 It takes approximately 2 seconds for sound to travel 686 meters through air. Based on this rate, what is the speed of sound in meters per second?

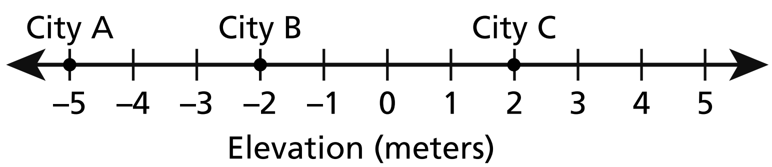
A 343

B 684

C 688

D 1,372

2 The elevations, in meters, of three cities as measured from sea level are marked on the number line shown below.



Based on the number line, which statement is true?

A City A and City B are both above sea level.

B City B is at sea level and City C is above sea level.

C City A is above sea level and City C is below sea level.

D City B is below sea level and City C is above sea level.

GO ON

3 Lacy needs to buy 72 ounces of beef for a recipe. Which expression can be used to determine how many pounds of beef Lacy needs for the recipe?

A $72 \div 16$

B 72×16

C $72 - 16$

D $72 + 16$

4 An expression is shown below.

$$2(12 - x) + y^2$$

What is the value of the expression when $x = 7$ and $y = 3$?

A 16

B 19

C 23

D 26

5 A rectangle is drawn on a coordinate plane with vertices located at $\mathrm{A}(-2, 3)$, $\mathrm{B}(4, 3)$, $\mathrm{C}(4, -7)$, and $\mathrm{D}(-2, -7)$. What is the length, in units, of side BC ?

A 6

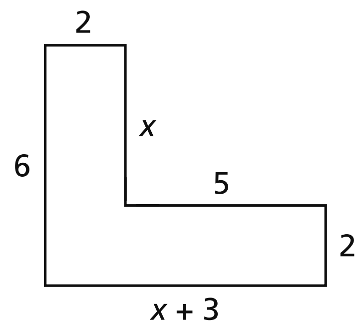
B 8

C 10

D 14

GO ON

6 Which expression represents the perimeter, in units, of the figure shown below?



A $x + 15$

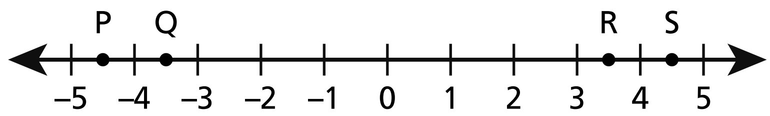
B $x + 18$

C $2x + 15$

D $2x + 18$

GO ON

8 Four points are plotted on the number line shown below.



Which point represents the opposite of $4\frac{1}{2}$?

A P

B Q

C R

D S

9 A teacher has $2\frac{1}{2}$ vanilla cakes. A serving size is $\frac{1}{12}$ of a cake. How many total servings of vanilla cake does the teacher have?

A 12

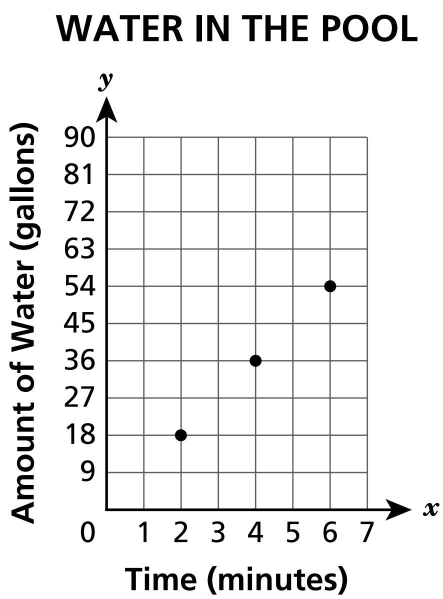
B 24

C 30

D 60

GO ON

10 A gardener is filling a pool with water from a garden hose at a constant rate. He plots points showing the number of gallons of water in the pool at different times on the coordinate plane shown below.



Based on the graph, what is the total amount of water, in gallons, in the pool at 3 minutes?

A 6

B 9

C 18

D 27

GO ON

11 An art teacher orders boxes to organize her supplies. Each box is a right rectangular prism with a length of $3\frac{2}{3}$ inches, a width of $3\frac{1}{2}$ inches, and a height of 6 inches. What is the volume, in cubic inches, of each box?

A $54\frac{1}{3}$

B $54\frac{3}{5}$

C 77

D 96

12 A flower garden has three different types of flowers. There are 12 daisies, 20 roses, and 13 tulips. What is the ratio of the number of roses to the total number of flowers?

A 20 : 25

B 20 : 45

C 25 : 20

D 45 : 20

GO ON

14 A pizza shop charges $\$15.00$ for each large pizza. There is a one time charge of $\$5.00$ to deliver any number of pizzas. Which equation could be used to determine the total charge, c, in dollars, of a delivery order of p large pizzas?

A $c = 15 + 5p$

B $c = 15p + 5$

C $c = 15 + 5 + p$

D $c = 15 \times 5 + p$

15 Marty needs to measure $\frac{3}{4}$ cup of cooking oil, but he can only find a tablespoon. He knows that 1 cup is the same as 16 tablespoons. How many tablespoons of oil will Marty need to measure $\frac{3}{4}$ cup of oil?

A 4

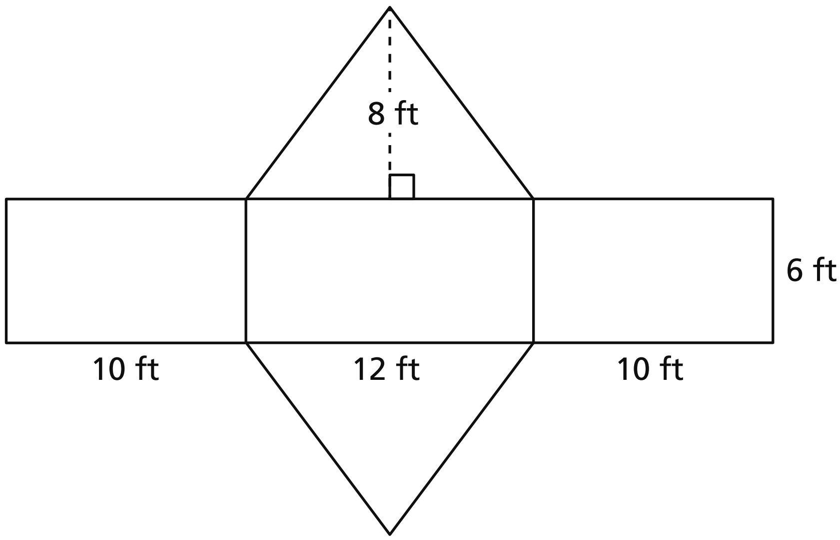
B 8

C 12

D 20

GO ON

16 A net of a right triangular prism is shown below.



What is the surface area, in square feet, of the triangular prism?

A 240

B 288

C 336

D 384

GO ON

17 A student has both basketball practice and choir practice today. Practices can occur any day of the week and follow the schedules shown below.

- basketball practice: every 6 days
- choir practice: every 8 days

What is the **fewest** number of days until the student has both basketball and choir practice on the same day again?

A 14

B 24

C 36

D 48

19 Which expression is equivalent to $5(6m + 3) + 4m$?

A $37m$

B $49m$

C $34m + 3$

D $34m + 15$

GO ON

20 Which statement is true for the values of -15 and -5 ?

A $-5 < -15$ and $|-5| < |-15|$

B $-5 > -15$ and $|-5| > |-15|$

C $-15 < -5$ and $|-15| > |-5|$

D $-15 > -5$ and $|-15| < |-5|$

21 Tina makes 6 paper decorations in 39 minutes. At this rate, how many minutes will it take Tina to make 14 paper decorations?

A 47

B 59

C 84

D 91

22 A model is built to represent a perfect cube using unit cubes. The model has 5 layers. Each layer consists of 5 rows of 5 unit cubes. What perfect cube is represented by the model?

A 125

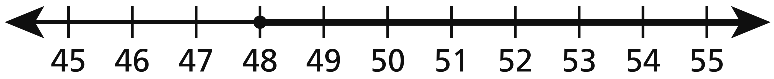
B 25

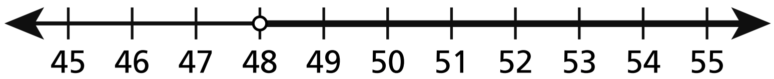
C 15

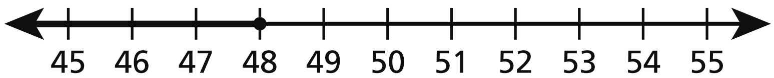
D 10

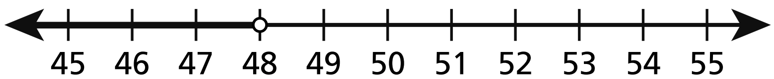
GO ON

23 At an amusement park, a rider must be at least 48 inches tall to ride a roller coaster. Which number line represents how tall, in inches, a rider must be to ride the roller coaster?

A 

B 

C 

D 

24 A soccer player scored 30% of the total goals scored by a team in a tournament. If the player scored 6 goals, how many total goals did the team score in the tournament?

A 14

B 18

C 20

D 24

GO ON

26 An inequality is shown below.

$$k - 16 > 29$$

Which value of k makes the inequality true?

A 30

B 35

C 45

D 58

27 A bakery sells boxes of cookies. The list below shows information for one month of sales.

- The bakery sold a total of 324 boxes of cookies.
- Of the boxes sold, 250 of the boxes contained just chocolate chip cookies.
- The rest of the boxes sold contained just sugar cookies.

Which equation can be used to determine the total number of boxes, x, of sugar cookies sold by the bakery that month?

A $\frac{324}{x} = 250$

B $324x = 250$

C $x - 250 = 324$

D $x + 250 = 324$

GO ON

28 A cake recipe requires 6 eggs for every 4 cups of flour. Which statement is true about the relationship between flour and eggs for the recipe?

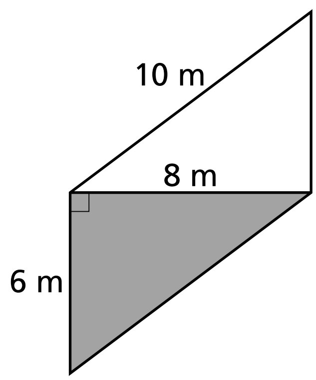
A For every 2 cups of flour, 3 eggs are used.

B For every 2 cups of flour, 5 eggs are used.

C For every 3 cups of flour, 2 eggs are used.

D For every 3 cups of flour, 5 eggs are used.

29 A piece of fabric in the shape of a parallelogram is used to create two sails for a sailboat. A diagram of the fabric is shown below.



What is the area, in square meters, of the shaded region?

A 24

B 30

C 48

D 60

GO ON

30 Which expression is equivalent to 55 less than the sum of 4 and twice a number, c ?

A $55 - 2(4 + c)$

B $55 - (4 + 2c)$

C $(4 + 2c) - 55$

D $2(4 + c) - 55$

STOP

Session 2

TIPS FOR TAKING THE TEST

Here are some ideas to help you do your best:

- Read each question carefully. Take your time.
- You have a ruler, a protractor, a reference sheet, and a calculator that you can use on the test if they help you answer the question.
- Be sure to show your work when asked.
- Be sure to explain your answer when asked.

31 A teacher surveyed all of the students in the art club to determine their favorite ice cream flavor. Each student chose one flavor. The results of the survey are shown in the table below.

SURVEY RESULTS

Favorite Ice Cream Flavor	Number of Students
Chocolate	18
Vanilla	21
Strawberry	11

What percent of the students in the art club chose vanilla ice cream as their favorite flavor?

A 21%

B 42%

C 58%

D 72%

32 An expression is shown below.

$$24x + 18y$$

Which expression is equivalent to the expression shown?

A $42xy$

B $4x + 3y$

C $6(4x + 3y)$

D $6(4x + 18y)$

GO ON

33 Points A and B are located 5 units away from each other on a coordinate plane. Which two ordered pairs could be the coordinates of points A and B ?

A $(-4, 1)$ and $(-4, -6)$

B $(1, -2)$ and $(1, 3)$

C $(-1, 2)$ and $(-4, 2)$

D $(3, -1)$ and $(2, -1)$

34 A store has a sale on one type of notebook. The price of 3 of these notebooks is $6. Which table shows the same relationship for each given number of notebooks of this type and the total price for those notebooks?

A

NOTEBOOK PRICES

Number of Notebooks	6	12	18	24	30
Price (dollars)	3	6	9	12	15

B

NOTEBOOK PRICES

Number of Notebooks	3	6	9	12	15
Price (dollars)	6	12	18	24	30

C

NOTEBOOK PRICES

Number of Notebooks	6	9	12	15	18
Price (dollars)	3	6	9	12	15

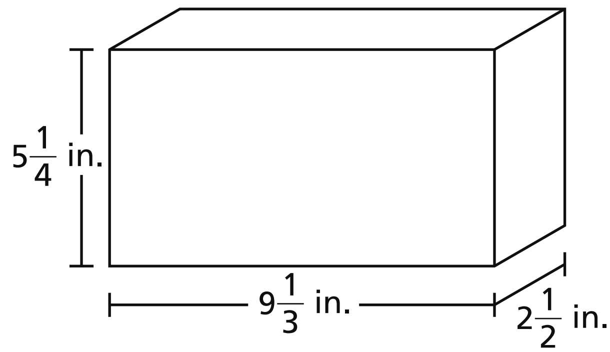
D

NOTEBOOK PRICES

Number of Notebooks	3	6	9	12	15
Price (dollars)	6	9	12	15	18

GO ON

35 A diagram of a right rectangular prism is shown below.



What is the volume, in cubic inches, of the right rectangular prism?

A $17\frac{1}{12}$

B $23\frac{1}{3}$

C $120\frac{1}{2}$

D $122\frac{1}{2}$

GO ON

38 **This question is worth 1 credit.**

What is the coefficient in the expression $6n^5$?

Answer ________________

GO ON

39 **This question is worth 1 credit.**

Write a numerical statement using >, <, or = that compares the two temperatures -5° Fahrenheit and -9° Fahrenheit.

Answer ______________________________

GO ON

41 **This question is worth 2 credits.**

Bella has a collection of 150 baseball cards. Dan has 43 fewer baseball cards in his collection than Bella has. Write and solve an equation to determine how many baseball cards, *c*, Dan has.

Show your work.

Answer __________________ cards

GO ON

42 **This question is worth 2 credits.**

The rules for creating two number patterns are shown below.

x: Start at 2, add 2

y: Start at 4, add 4

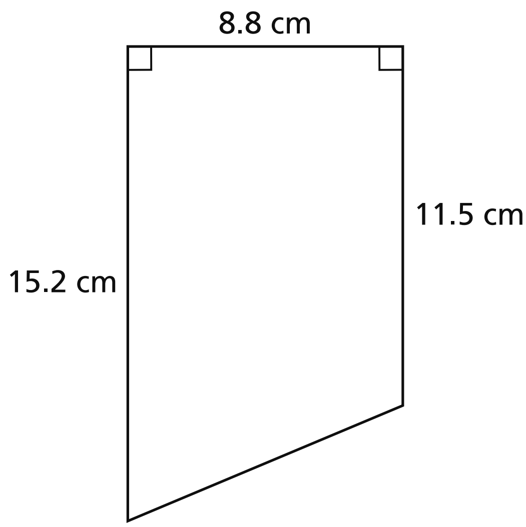
How do the corresponding values of x and y relate to each other? Be sure to include the first 5 numbers in both numerical patterns.

Explain your answer.

GO ON

43 **This question is worth 2 credits.**

A diagram of a trapezoid with dimensions in centimeters is shown below.



What is the area, in square centimeters, of the trapezoid?

Show your work.

Answer __________________ square centimeters

GO ON

44 **This question is worth 2 credits.**

Two stores sell the same size rolls of paper towels in different packages.

- Store A sells 6 rolls for $12.24
- Store B sells 9 rolls for $19.62

Which store has the lower price per roll of paper towels? Be sure to include the unit rate for each store in your answer.

Explain your answer.

GO ON

45 **This question is worth 2 credits.**

Evaluate the expression $3\left(4^3 - 6^2\right) + 7(2 + 1)^3$.

Show your work.

Answer ________________

GO ON

46 **This question is worth 3 credits.**

Jackie is making batches of trail mix based on the information listed below.

- She has $4\frac{3}{4}$ cups of dried bananas.
- She uses $\frac{1}{3}$ cup of dried bananas in each full batch of trail mix.

What is the maximum number of full batches of trail mix Jackie can make with the dried bananas she has?

Explain your answer.

STOP

THE STATE EDUCATION DEPARTMENT

THE UNIVERSITY OF THE STATE OF NEW YORK / ALBANY, NY 12234

2026 Mathematics Tests Map to the Standards

Grade 6

Question	Type	Key	Points	Standard	Cluster	Subscore	Secondary Standard(s)
Session 1							
1	Multiple Choice	A	1	NGLS.Math.Content.NY-6.RP.2	Ratios and Proportional Relationships	Ratios and Proportional Relationships	
2	Multiple Choice	D	1	NGLS.Math.Content.NY-6.NS.5	The Number System	The Number System	
3	Multiple Choice	A	1	NGLS.Math.Content.NY-6.RP.3d	Ratios and Proportional Relationships	Ratios and Proportional Relationships	
4	Multiple Choice	B	1	NGLS.Math.Content.NY-6.EE.2c	Expressions and Equations	Expressions and Equations	
5	Multiple Choice	C	1	NGLS.Math.Content.NY-6.G.3	Geometry		
6	Multiple Choice	D	1	NGLS.Math.Content.NY-6.EE.6	Expressions and Equations	Expressions and Equations	
8	Multiple Choice	A	1	NGLS.Math.Content.NY-6.NS.6a	The Number System	The Number System	
9	Multiple Choice	C	1	NGLS.Math.Content.NY-6.NS.1	The Number System	The Number System	
10	Multiple Choice	D	1	NGLS.Math.Content.NY-6.RP.3a	Ratios and Proportional Relationships	Ratios and Proportional Relationships	NGLS.Math.Content.NY-6.RP.3b
11	Multiple Choice	C	1	NGLS.Math.Content.NY-6.G.2	Geometry		
12	Multiple Choice	B	1	NGLS.Math.Content.NY-6.RP.1	Ratios and Proportional Relationships	Ratios and Proportional Relationships	
14	Multiple Choice	B	1	NGLS.Math.Content.NY-6.EE.9	Expressions and Equations	Expressions and Equations	
15	Multiple Choice	C	1	NGLS.Math.Content.NY-6.RP.3d	Ratios and Proportional Relationships	Ratios and Proportional Relationships	
16	Multiple Choice	B	1	NGLS.Math.Content.NY-6.G.4	Geometry		
17	Multiple Choice	B	1	NGLS.Math.Content.NY-6.NS.4	The Number System	The Number System	
19	Multiple Choice	D	1	NGLS.Math.Content.NY-6.EE.3	Expressions and Equations	Expressions and Equations	
20	Multiple Choice	C	1	NGLS.Math.Content.NY-6.NS.7d	The Number System	The Number System	
21	Multiple Choice	D	1	NGLS.Math.Content.NY-6.RP.3b	Ratios and Proportional Relationships	Ratios and Proportional Relationships	
22	Multiple Choice	A	1	NGLS.Math.Content.NY-6.G.5	Geometry		
23	Multiple Choice	A	1	NGLS.Math.Content.NY-6.EE.8	Expressions and Equations	Expressions and Equations	
24	Multiple Choice	C	1	NGLS.Math.Content.NY-6.RP.3c	Ratios and Proportional Relationships	Ratios and Proportional Relationships	
26	Multiple Choice	D	1	NGLS.Math.Content.NY-6.EE.5	Expressions and Equations	Expressions and Equations	
27	Multiple Choice	D	1	NGLS.Math.Content.NY-6.EE.7	Expressions and Equations	Expressions and Equations	
28	Multiple Choice	A	1	NGLS.Math.Content.NY-6.RP.1	Ratios and Proportional Relationships	Ratios and Proportional Relationships	
29	Multiple Choice	A	1	NGLS.Math.Content.NY-6.G.1	Geometry		
30	Multiple Choice	C	1	NGLS.Math.Content.NY-6.EE.2a	Expressions and Equations	Expressions and Equations	
Session 2							
31	Multiple Choice	B	1	NGLS.Math.Content.NY-6.RP.3c	Ratios and Proportional Relationships	Ratios and Proportional Relationships	
32	Multiple Choice	C	1	NGLS.Math.Content.NY-6.EE.4	Expressions and Equations	Expressions and Equations	
33	Multiple Choice	B	1	NGLS.Math.Content.NY-6.NS.8	The Number System	The Number System	
34	Multiple Choice	B	1	NGLS.Math.Content.NY-6.RP.3a	Ratios and Proportional Relationships	Ratios and Proportional Relationships	
35	Multiple Choice	D	1	NGLS.Math.Content.NY-6.G.2	Geometry		
38	Constructed Response		1	NGLS.Math.Content.NY-6.EE.2b	Expressions and Equations	Expressions and Equations	
39	Constructed Response		1	NGLS.Math.Content.NY-6.NS.7b	The Number System	The Number System	
41	Constructed Response		2	NGLS.Math.Content.NY-6.EE.7	Expressions and Equations	Expressions and Equations	
42	Constructed Response		2	NGLS.Math.Content.NY-5.OA.3	Expressions and Equations	Expressions and Equations	
43	Constructed Response		2	NGLS.Math.Content.NY-6.G.1	Geometry		
44	Constructed Response		2	NGLS.Math.Content.NY-6.RP.2	Ratios and Proportional Relationships	Ratios and Proportional Relationships	
45	Constructed Response		2	NGLS.Math.Content.NY-6.EE.1	Expressions and Equations	Expressions and Equations	
46	Constructed Response		3	NGLS.Math.Content.NY-6.NS.1	The Number System	The Number System	

*This item map is intended to identify the primary analytic skills necessary to successfully answer each question. However, some questions measure proficiencies described in multiple standards, including a balanced combination of procedural and conceptual understanding.

1-Credit Constructed-Response Rubric

1 Credit	A 1-credit response is a **correct answer** to the question which indicates a thorough understanding of mathematical concepts and/or procedures.
0 Credits*	A 0-credit response is incorrect, irrelevant, or incoherent.

* Condition Code A is applied whenever a student who is present for a test session leaves an entire constructed-response question in that session completely blank (no response attempted).

2-Credit Constructed-Response Holistic Rubric

2 Credits	A 2-credit response includes the correct solution to the question and demonstrates a thorough understanding of the mathematical concepts and/or procedures in the task. This response • indicates that the student has completed the task correctly, using mathematically sound procedures • contains sufficient work to demonstrate a thorough understanding of the mathematical concepts and/or procedures • may contain inconsequential errors that do not detract from the correct solution and the demonstration of a thorough understanding
1 Credit	A 1-credit response demonstrates only a partial understanding of the mathematical concepts and/or procedures in the task. This response • correctly addresses only some elements of the task • may contain an incorrect solution but applies a mathematically appropriate process • may contain the correct solution but required work is incomplete
0 Credits*	A 0-credit response is incorrect, irrelevant, incoherent, or contains a correct solution obtained using an obviously incorrect procedure. Although some elements may contain correct mathematical procedures, holistically they are not sufficient to demonstrate even a limited understanding of the mathematical concepts embodied in the task.

* Condition Code A is applied whenever a student who is present for a test session leaves an entire constructed-response question in that session completely blank (no response attempted).

3-Credit Constructed-Response Holistic Rubric

3 Credits	A 3-credit response includes the correct solution(s) to the question and demonstrates a thorough understanding of the mathematical concepts and/or procedures in the task. This response • indicates that the student has completed the task correctly, using mathematically sound procedures • contains sufficient work to demonstrate a thorough understanding of the mathematical concepts and/or procedures • may contain inconsequential errors that do not detract from the correct solution(s) and the demonstration of a thorough understanding
2 Credits	A 2-credit response demonstrates a partial understanding of the mathematical concepts and/or procedures in the task. This response • appropriately addresses most but not all aspects of the task using mathematically sound procedures • may contain an incorrect solution but provides sound procedures, reasoning, and/or explanations • may reflect some minor misunderstanding of the underlying mathematical concepts and/or procedures
1 Credit	A 1-credit response demonstrates only a limited understanding of the mathematical concepts and/or procedures in the task. This response • may address some elements of the task correctly but reaches an inadequate solution and/or provides reasoning that is faulty or incomplete • exhibits multiple flaws related to misunderstanding of important aspects of the task, misuse of mathematical procedures, or faulty mathematical reasoning • reflects a lack of essential understanding of the underlying mathematical concepts • may contain the correct solution(s) but required work is limited
0 Credits*	A 0-credit response is incorrect, irrelevant, incoherent, or contains a correct solution obtained using an obviously incorrect procedure. Although some elements may contain correct mathematical procedures, holistically they are not sufficient to demonstrate even a limited understanding of the mathematical concepts embodied in the task.

* Condition Code A is applied whenever a student who is present for a test session leaves an entire constructed-response question in that session completely blank (no response attempted).

1-Credit Constructed-Response Mathematics Scoring Policies

1. The student is **not** required to show work for a 1-credit constructed-response question, therefore, any work shown will **not** be scored. A clearly identified correct response should still receive full credit.
2. If the student clearly identifies a correct answer but fails to write that answer in the answer space, the student should still receive full credit.
3. If the student provides one legible response (and one response only), the rater should score the response, even if it has been crossed out.
4. If the student has written more than one response but has crossed some out, the rater should score only the response that has **not** been crossed out.
5. If the student provides more than one response but does not indicate which response is to be considered the correct response and none have been crossed out, the student shall not receive credit.
6. If the student does not provide the answer in the form as directed in the question, the student will not receive credit.
7. In questions requiring number sentences, the number sentences must be written horizontally.
8. When measuring angles with a protractor, there is a +/- 5 degrees deviation allowed of the true measure.
9. Condition Code A is applied whenever a student who is present for a test session leaves an entire constructed-response question in that session completely blank (no response attempted). This is not to be confused with a score of zero wherein the student does respond to part or all of the question, but that work results in a score of zero.

2- and 3-Credit Constructed-Response Mathematics Scoring Policies

1. If a student shows the work in other than a designated "Show your work" or "Explain" area, that work should still be scored.
2. If the question requires students to show their work, and the student shows appropriate work and clearly identifies a correct answer but fails to write that answer in the answer space, the student should still receive full credit.
3. If students are directed to show work or provide an explanation, a correct answer with **no** work shown or **no** explanation provided, receives **no** credit.
4. If students are **not** directed to show work, any work shown will **not** be scored. This applies to questions that do **not** ask for any work and questions that ask for work for one part and do **not** ask for work in another part.
5. If the student provides one legible response (and one response only), the rater should score the response, even if it has been crossed out.
6. If the student has written more than one response but has crossed some out, the rater should score only the response that has **not** been crossed out.
7. If the student provides more than one response, but does not indicate which response is to be considered the correct response and none have been crossed out, the student shall not receive full credit.
8. Trial-and-error responses are **not** subject to Scoring Policy #6 above, since crossing out is part of the trial-and-error process.
9. If a response shows repeated occurrences of the same conceptual error within a question, the conceptual error should **not** be considered more than once in gauging the demonstrated level of understanding.
10. In questions requiring number sentences, the number sentences must be written horizontally.
11. When measuring angles with a protractor, there is a +/- 5 degrees deviation allowed of the true measure.
12. Condition Code A is applied whenever a student who is present for a test session leaves an entire constructed-response question in that session completely blank (no response attempted). This is not to be confused with a score of zero wherein the student does respond to part or all of the question but that work results in a score of zero.

38

What is the coefficient in the expression $6n^5$?

Answer ______________

EXEMPLARY RESPONSE

38

What is the coefficient in the expression $6n^5$?

Answer 6

38

What is the coefficient in the expression $6n^5$?

Answer the coefficient for $6n^5$ is 6

Score Credit 1 (out of 1 credit)

A correct answer is provided.

38

What is the coefficient in the expression $6n^5$?

Answer 6

Score Credit 1 (out of 1 credit)

A correct answer is provided.

38

What is the coefficient in the expression $6n^5$?

Answer the coefficient in this expression is 6n

Score Credit 0 (out of 1 credit)

An incorrect answer is provided.

39

Write a numerical statement using >, <, or = that compares the two temperatures −5° Fahrenheit and −9° Fahrenheit.

Answer ______________________________

EXEMPLARY RESPONSE

39

Write a numerical statement using >, <, or = that compares the two temperatures −5° Fahrenheit and −9° Fahrenheit.

Answer $-9° < -5°$ *OR* $-5° > -9°$
OR other valid response

GUIDE PAPER 1

39

Write a numerical statement using >, <, or = that compares the two temperatures −5° Fahrenheit and −9° Fahrenheit.

Answer -5 > -9 Fahrenheit

Score Credit 1 (out of 1 credit)

A correct answer is provided.

39

Write a numerical statement using >, <, or = that compares the two temperatures −5° Fahrenheit and −9° Fahrenheit.

Answer $-9 < -5$

Score Credit 1 (out of 1 credit)

A correct answer is provided.

39

Write a numerical statement using >, <, or = that compares the two temperatures −5° Fahrenheit and −9° Fahrenheit.

Answer i say -5 is greater than -9 because -5 is closer to 0 but -9 is far away and is the bigger number but i say -5 < -9

Score Credit 0 (out of 1 credit)

An incorrect answer is provided.

41

Bella has a collection of 150 baseball cards. Dan has 43 fewer baseball cards in his collection than Bella has. Write and solve an equation to determine how many baseball cards, *c*, Dan has.

Show your work.

Answer ________________ cards

EXEMPLARY RESPONSE

41

Bella has a collection of 150 baseball cards. Dan has 43 fewer baseball cards in his collection than Bella has. Write and solve an equation to determine how many baseball cards, *c*, Dan has.

Show your work.

$$\begin{array}{r} 43 + c = 150 \\ \underline{-43 \quad = -43} \\ c = 107 \end{array}$$

OR other valid response

Answer 107 cards

41

Bella has a collection of 150 baseball cards. Dan has 43 fewer baseball cards in his collection than Bella has. Write and solve an equation to determine how many baseball cards, *c*, Dan has.

Show your work.

First I set up the equation:
$150 = c + 43$
Then I made sure the variable was by itself by subtracting 43
$- 43 = 0$
Then I had to do the same on the opposite side so I subtracted
$150 - 43 = 107$
So $c = 107$
Dan has 107 baseball cards

Answer 107 cards

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- An equation is provided that represents the situation using the given variable for the unknown number of cards that Dan has, and a mathematically sound procedure is used to determine the solution.

This response is complete and correct.

41

Bella has a collection of 150 baseball cards. Dan has 43 fewer baseball cards in his collection than Bella has. Write and solve an equation to determine how many baseball cards, *c*, Dan has.

Show your work.

c+43=150
150-43=107

Answer 107 cards

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- An equation is provided that represents the situation using the given variable for the unknown number of cards that Dan has, and a mathematically sound procedure is used to determine the solution.

This response is complete and correct.

41

Bella has a collection of 150 baseball cards. Dan has 43 fewer baseball cards in his collection than Bella has. Write and solve an equation to determine how many baseball cards, *c*, Dan has.

Show your work.

150-43=c

Answer 107 cards

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- An equation is provided that represents the situation using the given variable for the unknown number of cards that Dan has, and it is solved correctly.

This response contains sufficient work to show a thorough understanding.

GUIDE PAPER 4

41

Bella has a collection of 150 baseball cards. Dan has 43 fewer baseball cards in his collection than Bella has. Write and solve an equation to determine how many baseball cards, c, Dan has.

Show your work.

$$\begin{array}{r} 4\;10 \\ 1\cancel{5}\cancel{0} \\ -\;43 \\ \hline 107 \end{array}$$

Answer 107 cards

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- The solution for the total number of baseball cards Dan has is determined by a mathematically sound process; however, an equation that represents the situation using the given variable for the unknown is not provided.

This response contains the correct solution, but the required work is incomplete.

GUIDE PAPER 5

41

Bella has a collection of 150 baseball cards. Dan has 43 fewer baseball cards in his collection than Bella has. Write and solve an equation to determine how many baseball cards, c, Dan has.

Show your work.

$150 - c = 43$

$+150 \quad +150$

$c = 93$

Answer 93 cards

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- An equation that represents the situation using the given variable for the unknown number of cards that Dan has is provided; however, the equation is solved incorrectly.

This response correctly addresses only some elements of the task.

GUIDE PAPER 6

41

Bella has a collection of 150 baseball cards. Dan has 43 fewer baseball cards in his collection than Bella has. Write and solve an equation to determine how many baseball cards, *c*, Dan has.

Show your work.

$150 - 43 = x \qquad 150 - 43 = 3.4$

Answer 3.4 cards

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- An equation is provided that represents the situation using the variable *x*, instead of the given variable *c*, for the unknown number of cards that Dan has; however, a calculation error occurs when determining the solution.

This response correctly addresses only some elements of the task.

GUIDE PAPER 7

41

Bella has a collection of 150 baseball cards. Dan has 43 fewer baseball cards in his collection than Bella has. Write and solve an equation to determine how many baseball cards, *c*, Dan has.

Show your work.

$$\frac{\text{More}}{\text{Fewer}} = \frac{150}{43} = \frac{1}{c}$$

$$150\,c = 43$$

$$\overline{150c} \quad \overline{150c}$$

$$C = 3$$

Answer 3 cards

Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts and procedures in the task.

- An incorrect equation is written, and it is incorrectly solved.

This response is insufficient to show any understanding.

41

Bella has a collection of 150 baseball cards. Dan has 43 fewer baseball cards in his collection than Bella has. Write and solve an equation to determine how many baseball cards, *c*, Dan has.

Show your work.

Bella 150 cards. dan has 43 fewer cards than bella.
That means dan has 193 because 150+43=193

Answer Dan has 193. dan has 43 fewer cards than bella. cards

Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts and procedures in the task.

- The relationship between the number of cards that Dan has and the number of cards that Bella has is misinterpreted and used incorrectly to determine an incorrect solution.

Holistically, this response is insufficient to show any understanding.

42

The rules for creating two number patterns are shown below.

x: Start at 2, add 2

y: Start at 4, add 4

How do the corresponding values of x and y relate to each other? Be sure to include the first 5 numbers in both numerical patterns.

Explain your answer.

EXEMPLARY RESPONSE

42

The rules for creating two number patterns are shown below.

x: Start at 2, add 2

y: Start at 4, add 4

How do the corresponding values of x and y relate to each other? Be sure to include the first 5 numbers in both numerical patterns.

Explain your answer.

x: 2, 4, 6, 8, 10
y: 4, 8, 12, 16, 20
Each value of y is twice the value of the corresponding x.

OR other valid response

42

The rules for creating two number patterns are shown below.

x: Start at 2, add 2

y: Start at 4, add 4

How do the corresponding values of x and y relate to each other? Be sure to include the first 5 numbers in both numerical patterns.

Explain your answer.

x	y
2 +2	4 +4
4 +2	8 +4
6 +2	12 +4
8 +2	16 +4
10	20

Y is twice the amout of x

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The relationship between the corresponding values of x and y is stated and supported by sound reasoning that includes showing the first 5 numbers in both numerical patterns.

This response is complete and correct.

42

The rules for creating two number patterns are shown below.

x: Start at 2, add 2

y: Start at 4, add 4

How do the corresponding values of x and y relate to each other? Be sure to include the first 5 numbers in both numerical patterns.

Explain your answer.

The corresponding values of x and y relate to each other since the value of y is always 2 times greater than x.

Values of X: 2, 4, 6, 8, 10
Values of Y: 4, 8, 12, 16, 20

As you can see, X is always one half of Y, and Y is always two times greater than X.

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The relationship between the corresponding values of x and y is stated and supported by sound reasoning that includes showing the first 5 numbers in both numerical patterns.

This response is sufficient to demonstrate understanding.

42

The rules for creating two number patterns are shown below.

x: Start at 2, add 2

y: Start at 4, add 4

How do the corresponding values of x and y relate to each other? Be sure to include the first 5 numbers in both numerical patterns.

Explain your answer.

x=2,4,6,8,10
y=4,8,12,16,20
y is alaways 2 times x

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The relationship between the corresponding values of x and y is stated and supported by sound reasoning that includes showing the first 5 numbers in both numerical patterns.

This response is sufficient to demonstrate understanding.

GUIDE PAPER 4

42

The rules for creating two number patterns are shown below.

x: Start at 2, add 2

y: Start at 4, add 4

How do the corresponding values of x and y relate to each other? Be sure to include the first 5 numbers in both numerical patterns.

Explain your answer.

X and Y relate to eachother because x relates to y as in y is 4, 8, 12, 16, 20,... are all in x. Y relates to x as in take the first 5 numbers for instance and divide them by 2 to get the first 5 numbers of x.

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- Although a relationship between the corresponding values of x and y is stated, no additional reasoning is provided to show how or explain why pattern x generates numbers that support the stated relationship.

This response contains the correct solution but required work is incomplete.

42

The rules for creating two number patterns are shown below.

x: Start at 2, add 2

y: Start at 4, add 4

How do the corresponding values of x and y relate to each other? Be sure to include the first 5 numbers in both numerical patterns.

Explain your answer.

2 4 6 8 10
4 8 12 16 20 2 is half of 4 in
most multiples

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- The first 5 numbers in both numerical patterns are shown; however, the relationship between the corresponding values of x and y is not clearly stated, and insufficient reasoning is provided.

This response correctly addresses only some elements of the task.

42

The rules for creating two number patterns are shown below.

x: Start at 2, add 2

y: Start at 4, add 4

How do the corresponding values of x and y relate to each other? Be sure to include the first 5 numbers in both numerical patterns.

Explain your answer.

2,4,6,8,10,12
4,8,12,16,20
they would be simaler because they are adding the first nummber to the orriganal nuber and 2 add 2 will evenchuly get to 4 add4

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- The first 5 numbers in both numerical patterns are shown; however, insufficient reasoning is provided to support a relationship between the corresponding values of x and y.

This response correctly addresses only some elements of the task.

GUIDE PAPER 7

42

The rules for creating two number patterns are shown below.

x: Start at 2, add 2

y: Start at 4, add 4

How do the corresponding values of x and y relate to each other? Be sure to include the first 5 numbers in both numerical patterns.

Explain your answer.

2,4,6,8,10
4,8,16,20,24

Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts and procedures in the task.

- Although the first 5 numbers of pattern x are correct, a calculation error occurs when generating the first 5 numbers of pattern y and no relationship between corresponding values is stated or explained.

Holistically, this response is insufficient to demonstrate any understanding.

42

The rules for creating two number patterns are shown below.

x: Start at 2, add 2

y: Start at 4, add 4

How do the corresponding values of x and y relate to each other? Be sure to include the first 5 numbers in both numerical patterns.

Explain your answer.

2+ 2 = 4+ 4 = 8+ 8 = 16
4+ 4 = 8+ 8 = 16

they will both always equal the same when aded my the same numder

Score Credit 0 (out of 2 credits)

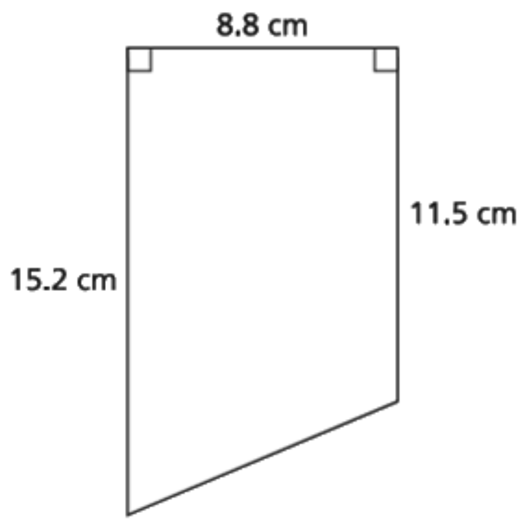
This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts and procedures in the task.

- The first 5 numbers of both numerical patterns are not included, and an incoherent explanation of a relationship is provided.

Holistically, this response is insufficient to demonstrate any understanding.

43

A diagram of a trapezoid with dimensions in centimeters is shown below.



What is the area, in square centimeters, of the trapezoid?

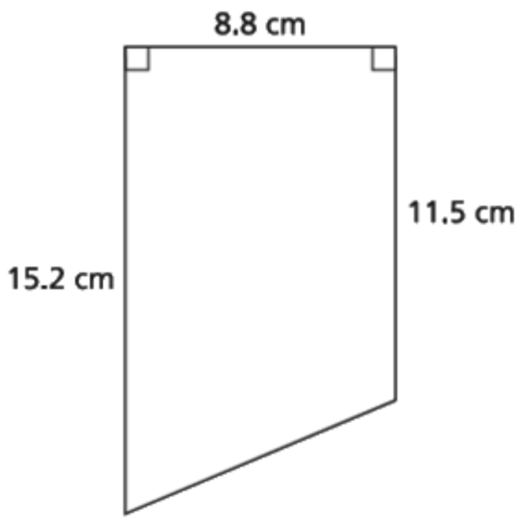
Show your work.

Answer ________________ square centimeters

EXEMPLARY RESPONSE

43

A diagram of a trapezoid with dimensions in centimeters is shown below.



What is the area, in square centimeters, of the trapezoid?

Show your work.

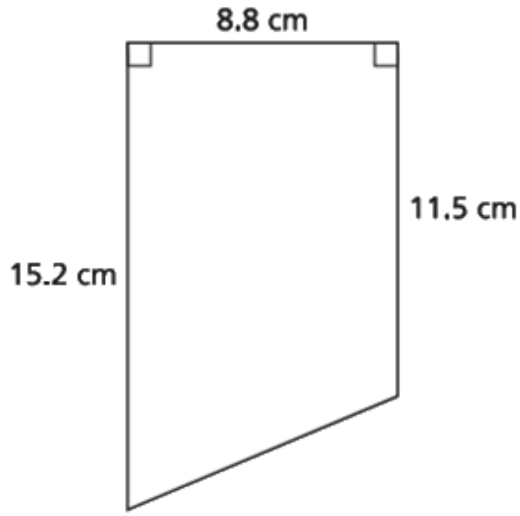
Area of rectangle:
$11.5 \times 8.8 = 101.2$
Area of triangle:
$15.2 - 11.5 = 3.7$
$(3.7 \times 8.8) \div 2 = 16.28$
Total: $101.2 + 16.28 = 117.48$ sq cm

OR other valid process

Answer 117.48 square centimeters

43

A diagram of a trapezoid with dimensions in centimeters is shown below.



What is the area, in square centimeters, of the trapezoid?

Show your work.

15.2-11.5=3.7
3.7 × 8.8 ÷ 2= 16.28

$$\begin{array}{r} 3.7 \\ \times\ 8.8 \\ \hline 32.56 \end{array}$$

11.5 × 8.8=101.2
101.2+16.28=117.48

Answer 117.48 square centimeters

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

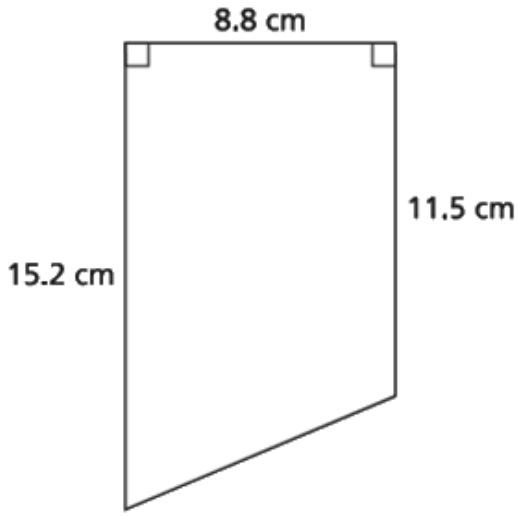
- The area of the trapezoid is determined and supported by sound procedures.

This response is complete and correct.

GUIDE PAPER 2

43

A diagram of a trapezoid with dimensions in centimeters is shown below.



What is the area, in square centimeters, of the trapezoid?

Show your work.

$15.2 - 11.5 = 3.7$
$3.7 \times 8.8 = 32.56$
$32.56 \div 2 = 16.28$
$15.2 \times 8.8 = 133.76$
$133.76 - 16.28 = 117.48$

Answer The trapizoid is 117.48 square centimeters

Score Credit 2 (out of 2 credits)

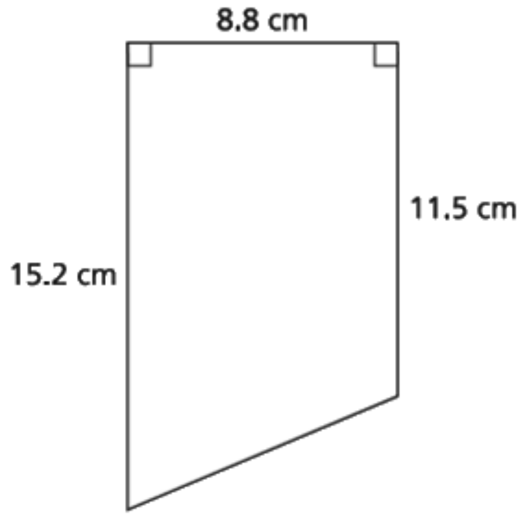
This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The area of the trapezoid is determined and supported by sound procedures.

This response is complete and correct.

43

A diagram of a trapezoid with dimensions in centimeters is shown below.



What is the area, in square centimeters, of the trapezoid?

Show your work.

15.2 + 11.5
26.7 × 4.4 = 117.48

Answer 117.48 square centimeters

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

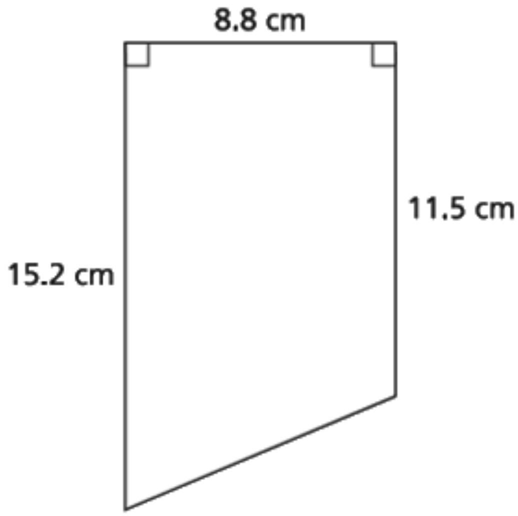
- The area of the trapezoid is determined and supported by a sound procedure.

This response contains sufficient work to show a thorough understanding.

GUIDE PAPER 4

43

A diagram of a trapezoid with dimensions in centimeters is shown below.



What is the area, in square centimeters, of the trapezoid?

Show your work.

15.2 − 11.5 =3.7
the rectangular part of the trapezoid is 11.5 × 8.8 which is 101.2
the triangular part of the trapezoid is 3.7 × 8.8 ÷ 2 which is 32.42
101.2 + 32.42=133.62

Answer 133.62 square centimeters

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

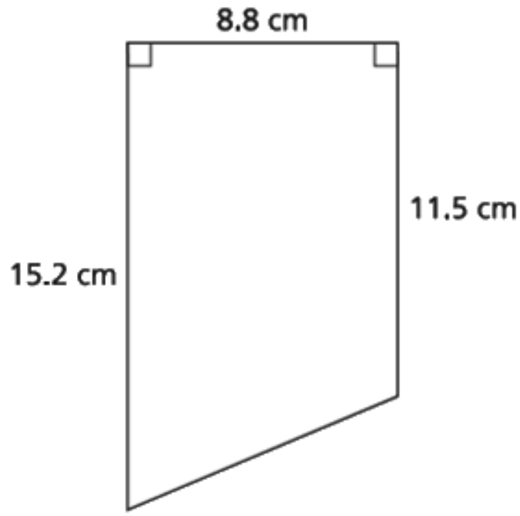
- The area of the trapezoid is determined using sound procedures; however, a calculation error occurs in computation of the area of the triangular part resulting in an incorrect solution.

This response contains an incorrect solution, but applies a mathematically appropriate process.

GUIDE PAPER 5

43

A diagram of a trapezoid with dimensions in centimeters is shown below.



What is the area, in square centimeters, of the trapezoid?

Show your work.

8.8x3.8= 32.56
8.8x11.2=101.2
101.2+32.56=133.76

Answer 133.76 square centimeters

Score Credit 1 (out of 2 credits)

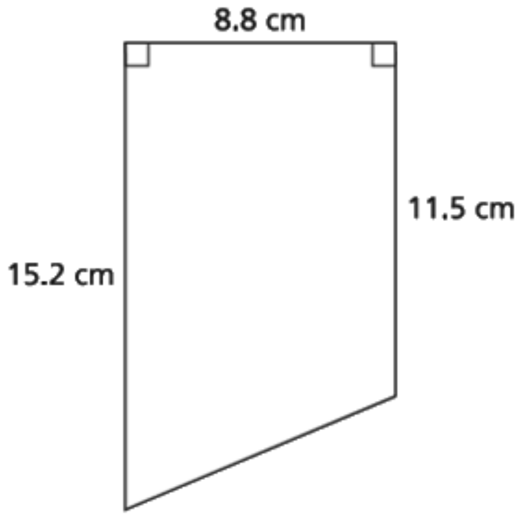
This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- Values are provided for an area decomposition of the trapezoid into two rectangles; however, the supporting work for those values is incorrect, and the need to use only half of the area of the smaller rectangle is not addressed, resulting in an incorrect solution.

This response correctly addresses only some elements of the task.

43

A diagram of a trapezoid with dimensions in centimeters is shown below.



What is the area, in square centimeters, of the trapezoid?

Show your work.

15.2+11.5=26.7 26.7 ÷ 2=13.35
13.35+8.8=22.15

Answer 22.15 square centimeters

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

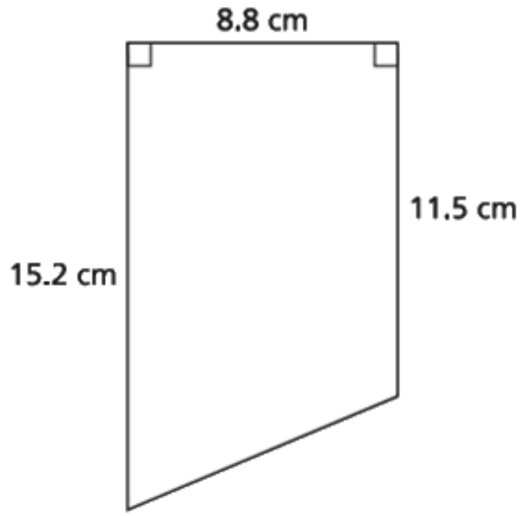
- Half of the sum of the two bases is inappropriately added to the height, instead of being multiplied, resulting in an incorrect solution.

This response correctly addresses only some elements of the task.

GUIDE PAPER 7

43

A diagram of a trapezoid with dimensions in centimeters is shown below.



What is the area, in square centimeters, of the trapezoid?

Show your work.

$15.2 \times 8.8 = 133.76$ $\quad\quad 133.76 \div 2 = 66.88$

Answer 66.88 square centimeters

Score Credit 0 (out of 2 credits)

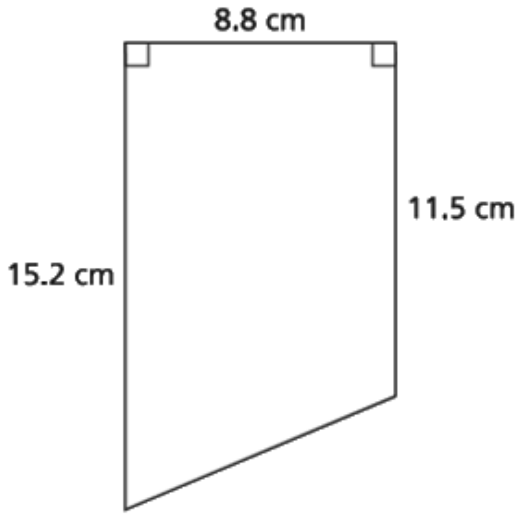
This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts and procedures in the task.

- Although a triangular part of the area of the trapezoid is correctly determined, it is incorrectly provided as the solution, and no further work is shown to address the rest of the composition of the area of the trapezoid.

Holistically, this response is insufficient to show any understanding.

43

A diagram of a trapezoid with dimensions in centimeters is shown below.



What is the area, in square centimeters, of the trapezoid?

Show your work.

(b1 × b2) + h ÷
(8.8 × 8.8) + 11.5 ÷ 2

Answer 83.19 square centimeters square centimeters

Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts and procedures in the task.

- An incorrect solution is obtained using an incorrect procedure.

This response is insufficient to show any understanding.

44

Two stores sell the same size rolls of paper towels in different packages.

- Store A sells 6 rolls for $12.24
- Store B sells 9 rolls for $19.62

Which store has the lower price per roll of paper towels? Be sure to include the unit rate for each store in your answer.

Explain your answer.

EXEMPLARY RESPONSE

44

Two stores sell the same size rolls of paper towels in different packages.

- Store A sells 6 rolls for \$12.24
- Store B sells 9 rolls for \$19.62

Which store has the lower price per roll of paper towels? Be sure to include the unit rate for each store in your answer.

Explain your answer.

Unit price for store A is $12.24 \div 6 = \$2.04$
Unit price for store B is $19.62 \div 9 = \$2.18$
Since $\$2.04 < \$2.18$,
store A has the lower price per roll of paper towels.

Or other valid explanation

GUIDE PAPER 1

44

Two stores sell the same size rolls of paper towels in different packages.

- Store A sells 6 rolls for $12.24
- Store B sells 9 rolls for $19.62

Which store has the lower price per roll of paper towels? Be sure to include the unit rate for each store in your answer.

Explain your answer.

I can solve and explain my answer by using unit rate. The cost for 6 rolls of T.P. from store A is $12.24. I can find out the cost of 1 roll by dividing 12.24 by 6 which equals 2.04. The cost for one roll of T.P. from store A is $2.04. I can find the unit rate of store B by dividing 19.62 by 9 to get 2.18. Store A's T.P. is cheaper because $2.04<2.18.

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The solution is stated and supported by sound reasoning that includes the calculation of the price per roll for each store.

This response is complete and correct.

GUIDE PAPER 2

44

Two stores sell the same size rolls of paper towels in different packages.

- Store A sells 6 rolls for \$12.24
- Store B sells 9 rolls for \$19.62

Which store has the lower price per roll of paper towels? Be sure to include the unit rate for each store in your answer.

Explain your answer.

Store A. 6 rolls : 12.24

Store B. 9 rolls : 19.62

A. 12.24 ÷ 6 = 2.04

B. 19.62 ÷ 9 = 2.18

Answer: Store A

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The solution is given and supported by the calculation of the price per roll for each store.

The response is sufficient to demonstrate understanding.

GUIDE PAPER 3

44

Two stores sell the same size rolls of paper towels in different packages.

- Store A sells 6 rolls for $12.24
- Store B sells 9 rolls for $19.62

Which store has the lower price per roll of paper towels? Be sure to include the unit rate for each store in your answer.

Explain your answer.

Store A has a lower price per roll of paper towels because store A is $2.04 and store B is $2.18. What I did was I divided the rolls by how many rolls they sell so store A had six rolls so I divided six by six to have one paper towel and I did the same to the money so I divided $12.24 by six and got $2.04. For store B I did the same thing I did 9 ÷ 9 to get one and I aslo did that to the money so I did $19.62 ÷ 9 and i got 2.18 and $2.04 is less then $2.18 so store A has a lower price.

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The solution is stated and supported by sound reasoning that includes the calculation of the price per roll for each store.

This response is complete and correct.

GUIDE PAPER 4

44

Two stores sell the same size rolls of paper towels in different packages.

- Store A sells 6 rolls for $12.24
- Store B sells 9 rolls for $19.62

Which store has the lower price per roll of paper towels? Be sure to include the unit rate for each store in your answer.

Explain your answer.

Store A: 1 Roll = $2.04
Store B: 1 Roll = $2.18
Store A is cheaper than Store B

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- Although the solution is stated, it is not clear how the supporting prices per roll are determined.

This response contains the correct solution, but the required work is incomplete.

GUIDE PAPER 5

44

Two stores sell the same size rolls of paper towels in different packages.

- Store A sells 6 rolls for \$12.24
- Store B sells 9 rolls for \$19.62

Which store has the lower price per roll of paper towels? Be sure to include the unit rate for each store in your answer.

Explain your answer.

store a sells for lower price because if you divde 12.24 by 6 it gives you 2.04 for each and store b you divde 19.62 by 6 and it gives youn 3.27 for each

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- A correct solution is stated based on the determined prices per roll; however, a divisor of 6 is incorrectly used instead of 9 for the calculation of the price per roll for store B.

This response correctly addresses only some elements of the task.

GUIDE PAPER 6

44

Two stores sell the same size rolls of paper towels in different packages.

- Store A sells 6 rolls for \$12.24
- Store B sells 9 rolls for \$19.62

Which store has the lower price per roll of paper towels? Be sure to include the unit rate for each store in your answer.

Explain your answer.

Store A	Store B
$6 \div 6 = 1$	$9 \div 9 = 1$
$12.24 \div 6 = 2.04$	$19.62 \div 9 = 2.18$

Store A has the lowest price per roll with $\frac{1}{2.04}$. Store B does not as Store B's unit rate is $\frac{1}{2.18}$.

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- Sound procedures are used to determine the price per roll for each store; however, faulty reasoning is then provided to support Store A having the lower price per roll due to the incorrect use of the reciprocal values of the prices per roll for both stores as the actual prices per roll.

This response correctly addresses only some elements of the task.

GUIDE PAPER 7

44

Two stores sell the same size rolls of paper towels in different packages.

- Store A sells 6 rolls for $12.24
- Store B sells 9 rolls for $19.62

Which store has the lower price per roll of paper towels? Be sure to include the unit rate for each store in your answer.

Explain your answer.

Store a has A lower price per roll because store A made it $2.03 per roll and store made it 3.48 per roll.

Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts and procedures in the task.

- Although store A is identified as the store with the lower price per roll, incorrect unit prices are stated, and no further supportive reasoning or work is provided.

This response is insufficient to demonstrate any understanding.

44

Two stores sell the same size rolls of paper towels in different packages.

- Store A sells 6 rolls for $12.24
- Store B sells 9 rolls for $19.62

Which store has the lower price per roll of paper towels? Be sure to include the unit rate for each store in your answer.

Explain your answer.

12.24 ÷ 12.24 = 1
19.62 ÷ 12.24 = 1.6 12.24 rolls are cheaper than the 19.62 rolls.

Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts and procedures in the task.

- Two division statements are written to show a relationship between the two given total prices, and no further explanation surrounding the price per roll at each store is provided.

This response is insufficient to demonstrate any understanding.

45

Evaluate the expression $3(4^3 - 6^2) + 7(2 + 1)^3$.

Show your work.

Answer ________________

EXEMPLARY RESPONSE

45

Evaluate the expression $3(4^3 - 6^2) + 7(2 + 1)^3$.

Show your work.

$3(4^3 - 6^2) + 7(2 + 1)^3 =$
$3(64 - 36) + 7(3)^3 =$
$3(28) + 7(27)$
$= 84 + 189$
$= 273$

OR other valid process

Answer 273

45

Evaluate the expression $3(4^3 - 6^2) + 7(2 + 1)^3$.

Show your work.

$3\left(4^3 - 6^2\right) + 7(2 + 1)^3$

$3\left(64 - 36\right) + 7(3)^3$

$3\left(28\right) + 7\left(27\right)$

$84+189$

273

Answer 273

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The value of the expression is determined using sound procedures.

This response is complete and correct.

GUIDE PAPER 2

45

Evaluate the expression $3(4^3 - 6^2) + 7(2+1)^3$.

Show your work.

First I multipied 4 by 4 by 4 to get a answer of 64, then I multiplied 6 by 6 to get 36, then I had to subtract to get 28 then I multiplied that by 3 to get 84. Then I went to my next problems. I added 2 with 1 to get 3 to then multiply by 3 and 3, to get a answer of 27 so then I multiplied that by 7 to get 189, then I had to add 189 to 84 to get my answer.

Answer 273

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The value of the expression is determined using sound procedures.

This response is complete and correct.

45

Evaluate the expression $3(4^3 - 6^2) + 7(2 + 1)^3$.

Show your work.

84 left
189 right
$3(28) + 7(3)^3$
$3(28) + 7(27)$
$84 + 189$

Answer 273

Score Credit 2 (out of 2 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The value of the expression is determined using sound procedures.

This response contains sufficient work to show a thorough understanding.

GUIDE PAPER 4

45

Evaluate the expression $3(4^3 - 6^2) + 7(2 + 1)^3$.

Show your work.

$3(4^3-6^2)$

$(64-36)$

3×28

84

$7(2+1)^3$

$(3)^3$

$7(9=63$

$$\begin{array}{r} 63 \\ +\ 84 \\ \hline 147 \end{array}$$

Answer 147

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- Sound procedures are used to determine the value of the expression; however, a calculation error when evaluating 3^3 results in an incorrect solution.

This response contains an incorrect solution, but applies a mathematically appropriate process.

GUIDE PAPER 5

45

Evaluate the expression $3(4^3 - 6^2) + 7(2 + 1)^3$.

Show your work.

4 to the third power = 64, 64 - 6 squared, 6 squared = 36. 64 - 36 = 28. 3 × 28 = 84.
2 + 1 = 3, 3 to the third power equals 27, 27 + 7 = 34 .

So, 84 + 34 = 118

Answer 118

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- An incorrect value of the expression is determined due to 27 being inappropriately added to 7 instead of being multiplied; however, the rest of the work needed to evaluate the expression is carried out correctly using sound procedures.

This response correctly addresses only some elements of the task.

GUIDE PAPER 6

45

Evaluate the expression $3(4^3 - 6^2) + 7(2+1)^3$.

Show your work.

3(64-36) 7 x 27
192-36 = 156 = 189

156+189 = 345

The expression equal 345.

Answer 345

Score Credit 1 (out of 2 credits)

This response demonstrates only a partial understanding of the mathematical concepts and procedures in the task.

- Although an incorrect value for the expression is determined due to the incorrect distribution of the 3, the rest of work needed to evaluate the expression is carried out correctly using sound procedures.

This response correctly addresses only some elements of the task.

45

Evaluate the expression $3(4^3 - 6^2) + 7(2+1)^3$.

Show your work.

4x4x4=64
64x3=192
192-36=156
156+21=177 6x6=36
7x3=21

Answer 177

Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts and procedures in the task.

- Although this response contains some correct elements, incorrect procedures are used to evaluate the expression.

Holistically, this response is insufficient to show any understanding.

GUIDE PAPER 8

45

Evaluate the expression $3(4^3 - 6^2) + 7(2+1)^3$.

Show your work.

$3(4^3 - 6^2) + 7(2+1)^3 = ?$

Answer 273

Score Credit 0 (out of 2 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts and procedures in the task.

- The correct value of the expression is provided with no work.

Per Scoring Policy #3 for 2- and 3-credit responses, this response receives no credit.

46

Jackie is making batches of trail mix based on the information listed below.

- She has $4\frac{3}{4}$ cups of dried bananas.
- She uses $\frac{1}{3}$ cup of dried bananas in each full batch of trail mix.

What is the maximum number of full batches of trail mix Jackie can make with the dried bananas she has?

Explain your answer.

EXEMPLARY RESPONSE

46

Jackie is making batches of trail mix based on the information listed below.

- She has $4\frac{3}{4}$ cups of dried bananas.
- She uses $\frac{1}{3}$ cup of dried bananas in each full batch of trail mix.

What is the maximum number of full batches of trail mix Jackie can make with the dried bananas she has?

Explain your answer.

$4^3/_4 \div {}^1/_3 = {}^{19}/_4 \div {}^1/_3 = {}^{19}/_4 \times {}^3/_1 = {}^{57}/_4 = 14^1/_4$

Since she can't make a full batch out of the part left, she only has enough for 14 full batches.

OR other valid explanation

46

Jackie is making batches of trail mix based on the information listed below.

- She has $4\frac{3}{4}$ cups of dried bananas.
- She uses $\frac{1}{3}$ cup of dried bananas in each full batch of trail mix.

What is the maximum number of full batches of trail mix Jackie can make with the dried bananas she has?

Explain your answer.

$$4\frac{3}{4} = \frac{19}{4} \quad \frac{19}{4} \div \frac{1}{3} = \frac{19}{4} \times \frac{3}{1} = \frac{57}{4} = 14\frac{1}{4}$$

So Jackie can make 14 full batches of trail mix.

Score Credit 3 (out of 3 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The number of batches of trail mix that can be made with the bananas is correctly determined using a sound procedure and interpreted correctly for the final solution.

This response is complete and correct.

46

Jackie is making batches of trail mix based on the information listed below.

- She has $4\frac{3}{4}$ cups of dried bananas.
- She uses $\frac{1}{3}$ cup of dried bananas in each full batch of trail mix.

What is the maximum number of full batches of trail mix Jackie can make with the dried bananas she has?

Explain your answer.

The maximum number is 14 batches

$$\frac{19}{4} \times \frac{3}{1} = \frac{57}{4} = 14.25$$

Score Credit 3 (out of 3 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The number of batches of trail mix that can be made with the bananas is correctly determined using a sound procedure and interpreted correctly for the final solution.

This response is sufficient to demonstrate a thorough understanding.

46

Jackie is making batches of trail mix based on the information listed below.

- She has $4\frac{3}{4}$ cups of dried bananas.
- She uses $\frac{1}{3}$ cup of dried bananas in each full batch of trail mix.

What is the maximum number of full batches of trail mix Jackie can make with the dried bananas she has?

Explain your answer.

$$4\frac{3}{4} = \frac{19}{4}$$

$$\frac{19}{4} = \frac{57}{12}$$

$$\frac{1}{3} = \frac{4}{12}$$

$57 \div 4 = 14.25$

jackie can make 14 full batches of trail mix

Score Credit 3 (out of 3 credits)

This response demonstrates a thorough understanding of the mathematical concepts and procedures in the task.

- The number of batches of trail mix that can be made with the bananas is correctly determined using a sound procedure and interpreted correctly for the final solution.

This response is complete and correct.

46

Jackie is making batches of trail mix based on the information listed below.

- She has $4\frac{3}{4}$ cups of dried bananas.
- She uses $\frac{1}{3}$ cup of dried bananas in each full batch of trail mix.

What is the maximum number of full batches of trail mix Jackie can make with the dried bananas she has?

Explain your answer.

The maximum number of full batches of trail mix jackie can make with the dried bananas she has is 14 because you have to divide $4\frac{3}{4}$ by $\frac{1}{3}$ to see how many times $\frac{1}{3}$ can go into $4\frac{3}{4}$. which is $14\frac{1}{4}$ but in full batches it would be 14.

Score Credit 2 (out of 3 credits)

This response demonstrates a partial understanding of the mathematical concepts and procedures in the task.

- The number of batches of trail mix Jackie can make with the bananas is determined and interpreted correctly for the final solution; however, the explanation as to how the division results in $14\frac{1}{4}$ is insufficient.

This response appropriately addresses most, but not all, aspects of the task.

GUIDE PAPER 5

46

Jackie is making batches of trail mix based on the information listed below.

- **She has $4\frac{3}{4}$ cups of dried bananas.**
- **She uses $\frac{1}{3}$ cup of dried bananas in each full batch of trail mix.**

What is the maximum number of full batches of trail mix Jackie can make with the dried bananas she has?

Explain your answer.

$$4\frac{3}{4} \cdot \cancel{\frac{1}{3}} \left(\frac{3}{1}\right)$$

$$\frac{19}{4} \cdot \frac{3}{1} = \frac{57}{4}$$

$$\boxed{14.25}$$

$$\begin{array}{r} 14.25 \\ 4\overline{)57.00} \\ -4 \\ 17 \\ -16 \\ 100 \\ 100 \end{array}$$

Score Credit 2 (out of 3 credits)

This response demonstrates a partial understanding of the mathematical concepts and procedures in the task.

- The number of batches of trail mix Jackie can make with the bananas is determined using a sound procedure; however, this number is not interpreted and is incorrectly provided as the final solution.

This response appropriately addresses most, but not all, aspects of the task.

GUIDE PAPER 6

46

Jackie is making batches of trail mix based on the information listed below.

- She has $4\frac{3}{4}$ cups of dried bananas.
- She uses $\frac{1}{3}$ cup of dried bananas in each full batch of trail mix.

What is the maximum number of full batches of trail mix Jackie can make with the dried bananas she has?

Explain your answer.

Jakie can make 14 full batches with the dried bananas she has because when you turn $4\frac{3}{4}$ into an improper fraction, and then make $\frac{1}{3}$ and $\frac{19}{4}$ have a same denominator, $\frac{4}{12}$ can go into $\frac{57}{12}$ 14 times smoothly.

Score Credit 2 (out of 3 credits)

This response demonstrates a partial understanding of the mathematical concepts and procedures in the task.

- The maximum number of full batches of trail mix Jackie can make with the bananas is given; however, the stated reasoning does not fully support the final solution due to the lack of clearly addressing the fractional remainder of ¼ that results from the division.

This response appropriately addresses most, but not all, aspects of the task.

GUIDE PAPER 7

46

Jackie is making batches of trail mix based on the information listed below.

- She has $4\frac{3}{4}$ cups of dried bananas.
- She uses $\frac{1}{3}$ cup of dried bananas in each full batch of trail mix.

What is the maximum number of full batches of trail mix Jackie can make with the dried bananas she has?

Explain your answer.

$$4\frac{3}{4} \div \frac{1}{3} = 14\frac{1}{4}$$

Score Credit 1 (out of 3 credits)

This response demonstrates only a limited understanding of the mathematical concepts and procedures in the task.

- A correct equation is written that shows a process that can be used to determine the number of batches of trail mix Jackie can make with the bananas; however, no further explanation or work is provided to support the value of $14\frac{1}{4}$.
- The number of batches of trail mix Jackie can make is not interpreted and incorrectly provided as the final solution.

This response addresses some elements of the task correctly but reaches an inadequate solution and provides reasoning that is faulty or incomplete.

46

Jackie is making batches of trail mix based on the information listed below.

- She has $4\frac{3}{4}$ cups of dried bananas.
- She uses $\frac{1}{3}$ cup of dried bananas in each full batch of trail mix.

What is the maximum number of full batches of trail mix Jackie can make with the dried bananas she has?

Explain your answer.

She can make 4 9/4 full batches

$4\frac{3}{4} \times \frac{3}{1} = 4\frac{9}{4}$

Score Credit 1 (out of 3 credits)

This response demonstrates only a limited understanding of the mathematical concepts and procedures in the task.

- A correct multiplication process is shown in the equation written that can be used to determine the number of batches of trail mix Jackie can make with the bananas; however, the multiplication is performed incorrectly.
- The determined number of batches of trail mix Jackie can make is not interpreted and incorrectly provided as the final solution.

This response addresses some elements of the task correctly but reaches an inadequate solution and provides reasoning that is faulty or incomplete.

GUIDE PAPER 9

46

Jackie is making batches of trail mix based on the information listed below.

- She has $4\frac{3}{4}$ cups of dried bananas.
- She uses $\frac{1}{3}$ cup of dried bananas in each full batch of trail mix.

What is the maximum number of full batches of trail mix Jackie can make with the dried bananas she has?

Explain your answer.

its 14 because if you do the amount she has divevided by how much batches she uses it equals 14

Score Credit 1 (out of 3 credits)

This response demonstrates only a limited understanding of the mathematical concepts and procedures in the task.

- A sound division process is described that can be used to determine the number of batches of trail mix Jackie can make with the bananas; however, no further explanation or work of the division process is provided.
- Although the correct solution is given, the division process described does not fully support the solution due to the lack of addressing the fractional remainder of ¼.

This response contains the correct solution, but the required work is limited.

GUIDE PAPER 10

46

Jackie is making batches of trail mix based on the information listed below.

- She has $4\frac{3}{4}$ cups of dried bananas.
- She uses $\frac{1}{3}$ cup of dried bananas in each full batch of trail mix.

What is the maximum number of full batches of trail mix Jackie can make with the dried bananas she has?

Explain your answer.

$4\frac{3}{4} - \frac{1}{3} =$

$\frac{15}{4} - \frac{1}{3} = \frac{14}{1} = 14$ trail mix

Score Credit 0 (out of 3 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts and procedures in the task.

- The correct solution is obtained using an obviously incorrect procedure.

This response is insufficient to demonstrate any understanding.

46

Jackie is making batches of trail mix based on the information listed below.

- She has $4\frac{3}{4}$ cups of dried bananas.
- She uses $\frac{1}{3}$ cup of dried bananas in each full batch of trail mix.

What is the maximum number of full batches of trail mix Jackie can make with the dried bananas she has?

Explain your answer.

14 batches of trail mix.

Score Credit 0 (out of 3 credits)

This response is not sufficient to demonstrate even a limited understanding of the mathematical concepts and procedures in the task.

- The correct solution is provided with no explanation.

Per Scoring Policy #3 for 2- and 3-credit responses, this response receives no credit.